improve_qr

take y s.t. $\Sigma(x_i x_j + y^2)^2$ has min

$y^2 \sim \frac{1}{n(n+1)}$, $y \sim \frac{1}{n}$

執行：① 分 n 群

② 捨棄前 i 個 eigenvector

③ 取 $i+1 \sim i+(n-1)$ 個 eigenvector

④ 用 $(n-1)$個 + $(\frac{1}{n} \sim \frac{2}{n})$

做 cluster_qr (n：data 數)

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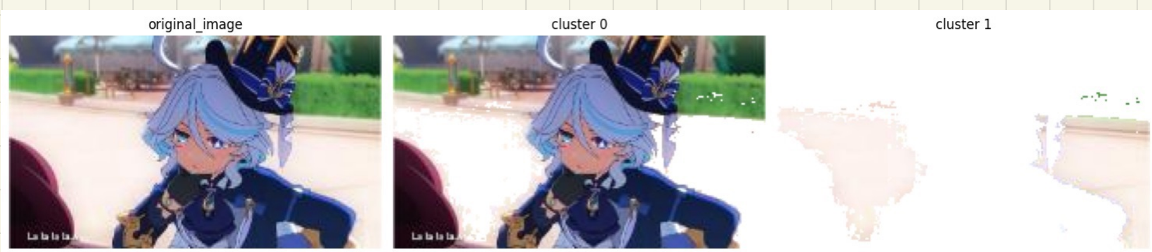
result of improve_qr

test1 (2 cluster)

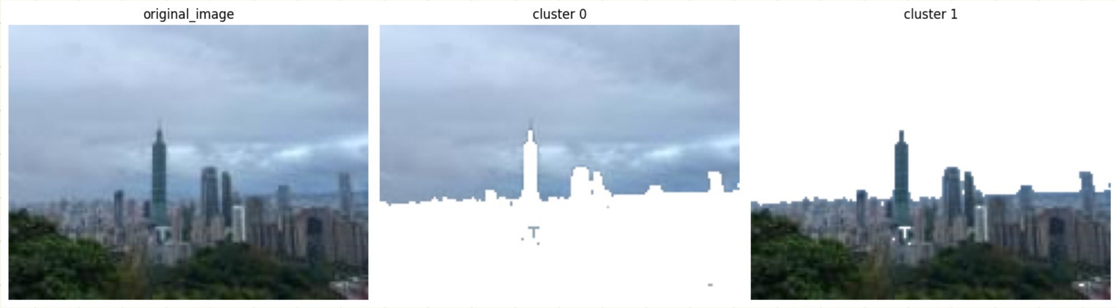


test 1 (3 cluster)

test 2 (2 cluster)

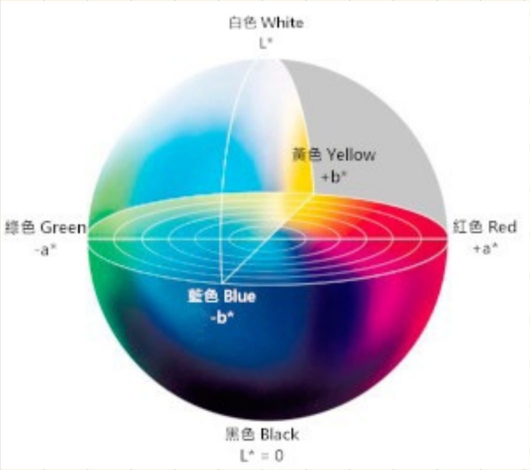
---

kmean v.s. gr



CIELAB (Lab)

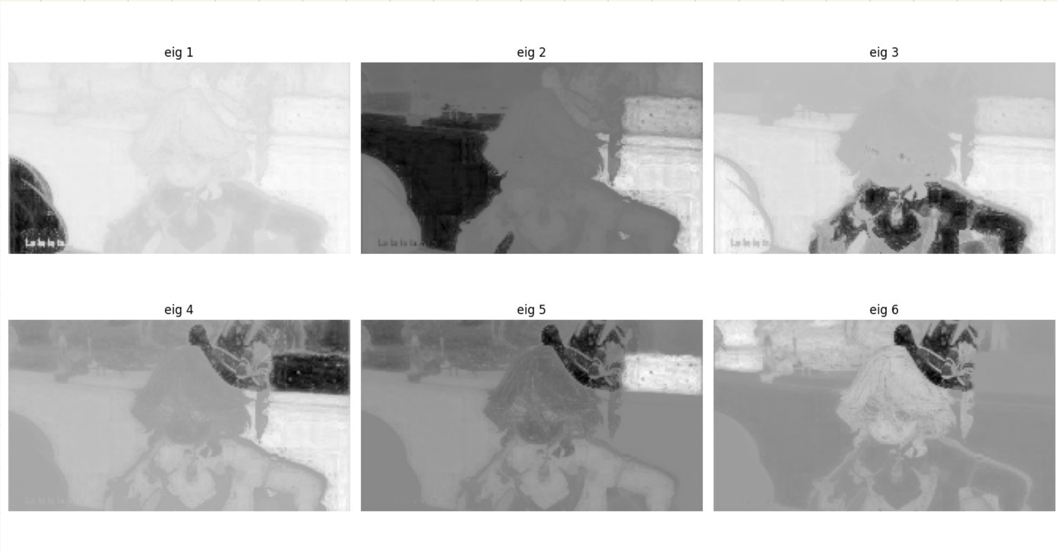
(L, a, b) ⇒ { L：亮度 (0～100)
a：色度(紅綠)
b：色度 (黃藍)



(L, a, b, x, y)，正規化到 (0, 1)

result :

RGB:



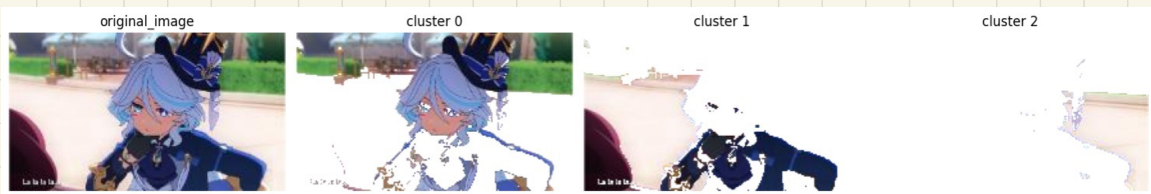
Lab :

kmeans , 2 cluster

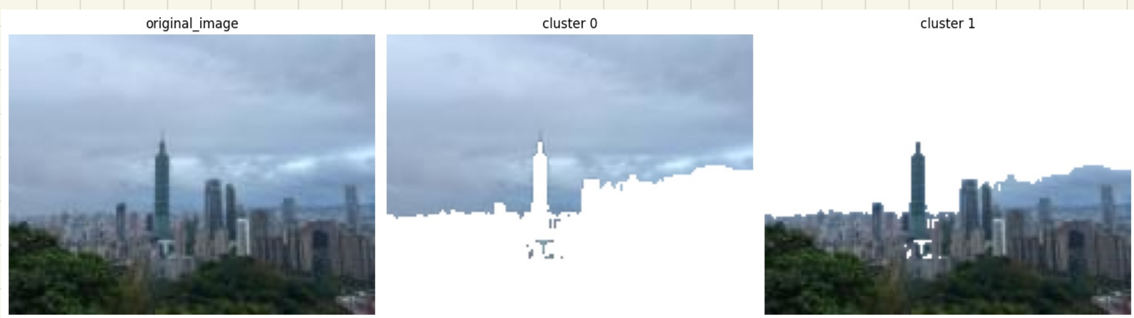
kmeans , 3 cluster

vs.

kmeans , 3 cluster (RGB)

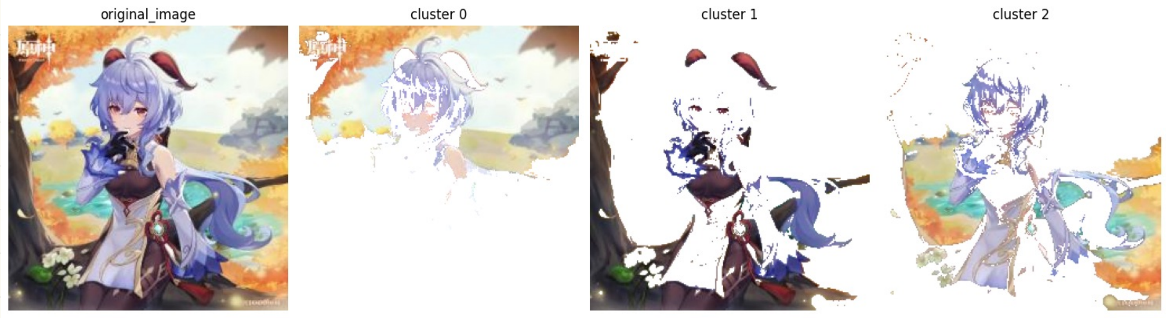


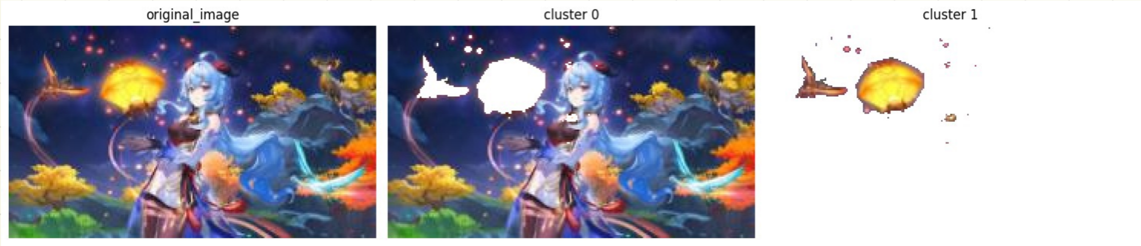
test 2 , kmean , 2 cluster

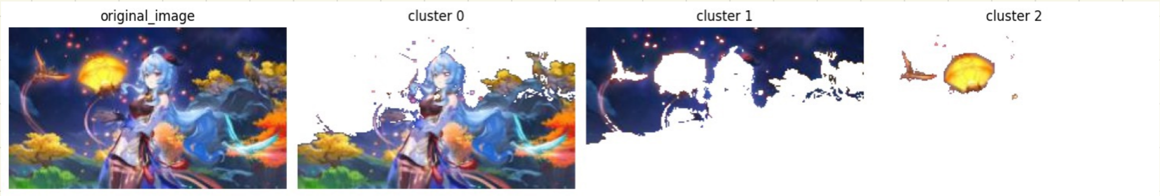


other_test

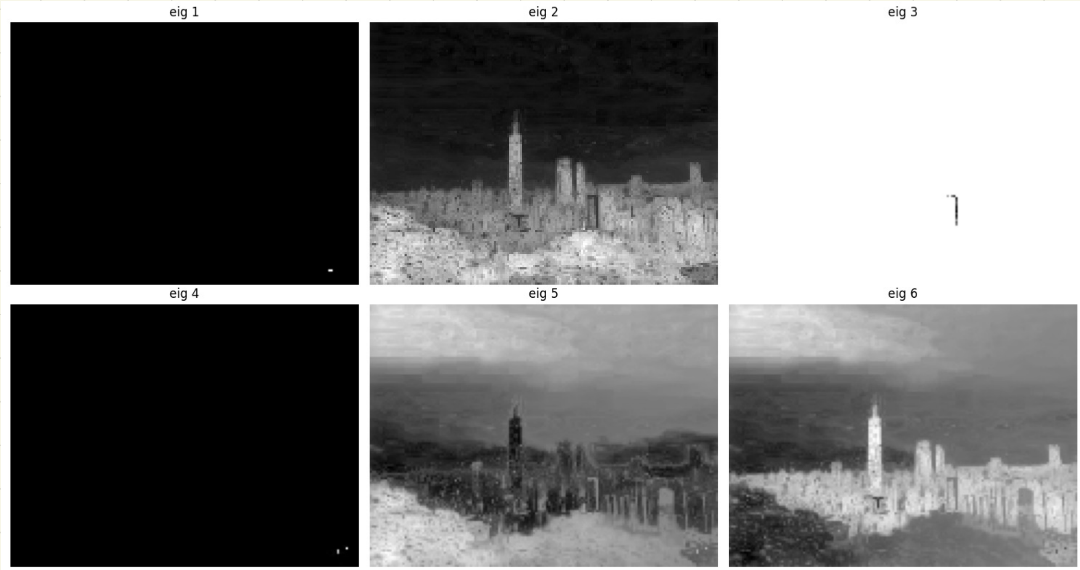

original_image
cluster 0
cluster 1


original_image
cluster 0
cluster 1
cluster 2

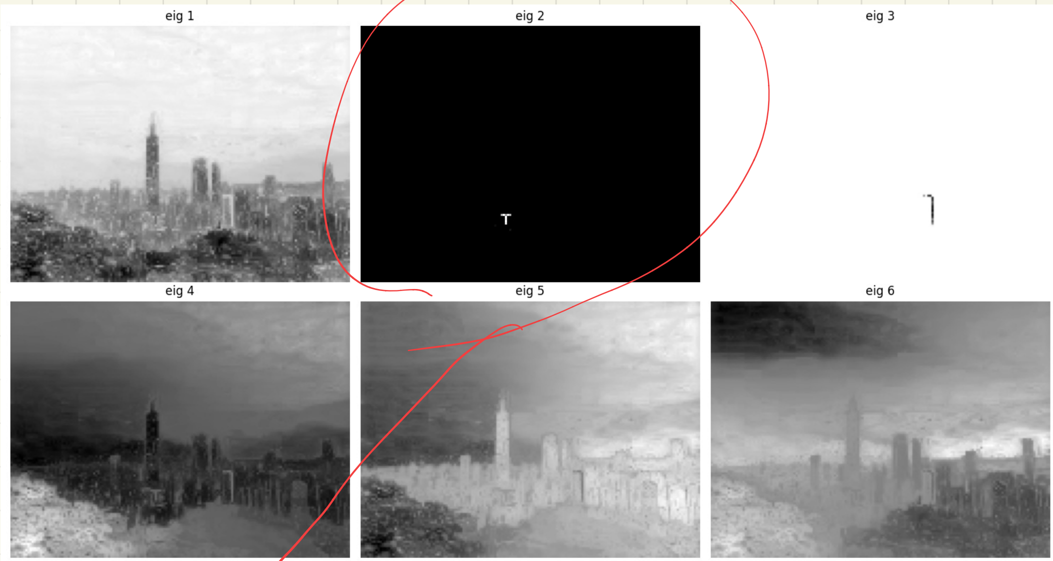

original_image
cluster 0
cluster 1

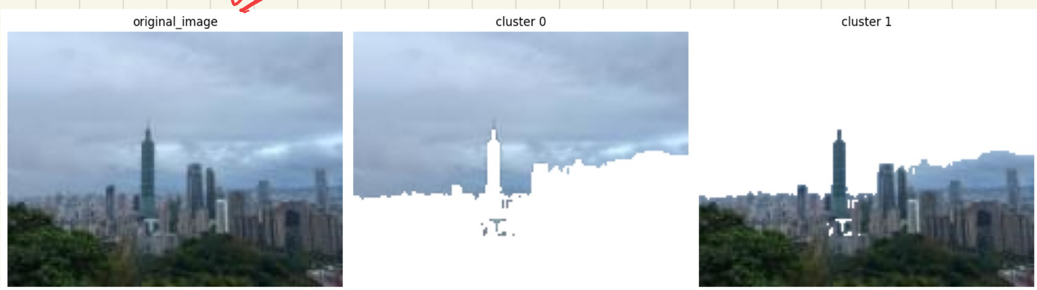

original_image
cluster 0
cluster 1
cluster 2

(ii) eigenvector 差異



(2) 怎麼分的


eig 1
eig 2
eig 3
eig 4
eig 5
eig 6


original_image
cluster 0
cluster 1

(3) qr 有用?